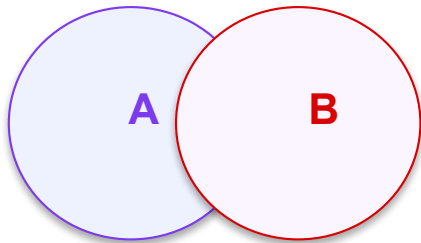


01 Sets (Set Concepts)

Mathematics



Mathematics

Look • wonder • predict

OPENING QUESTION

**Big question: How many stones
are in my hand?**

First think alone. Then tell a partner one possible answer.

Today's mission

Goal: I can distinguish between finite and infinite sets using

1 Learn

Use: Can We Count Everything?, Defining Our Sets

2 Practise

Show: answer with one reason or example.

3 Show

Routine: listen, talk, try, explain.

4 Reflect

Explain in your own words.



Starter: think first

1

THINK

**How many stones are
in my hand?**

30 seconds

2

PAIR

**compare your first
answer with a partner.**

with partner

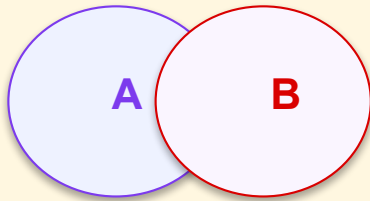
3

SHARE

**connect your answer
to Can We Count
Everything?.**

whole class

Can We Count Everything?



Mathematics

KEY IDEA

Key idea: Concrete exploration of countable objects (stones) vs. uncountable

1

Practical counting of stones.

2

Attempting to count grains of sand.

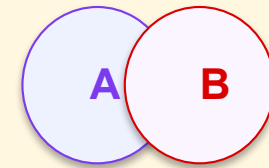
3

Identifying which count can be finished.

Defining Our Sets

BOARD MODEL

1. Finite Set: Has a countable number of members.
2. Infinite Set: Has members that go on forever.
3. Examples from daily life.



Mathematics

**Teacher models first.
Learners copy the
pattern, then explain
the step.**

Try it together

Answer with evidence:



Can you tell me exactly how many grains are in that pile of sand?



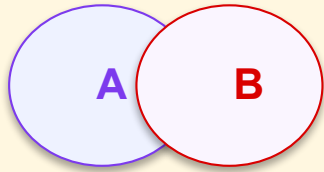
Which count was easier to finish?



What is a finite set?

Use the source idea, not guessing.

Activity challenge



Mathematics

GROUP CHALLENGE

Produce one answer the class can check.

1

In groups of five, learners count a pile of 10 bottle caps and then try to count

2

Write the definitions of finite and infinite sets in their exercise books and list

3

Draw two Venn diagrams. In the first, they represent

Class share-out: one group explains, one group improves it.

Exit ticket

EXIT TICKET

Answer: Can a very large set still be finite? (Yes, if it has an end).

Use one complete sentence.

Use one key word: Set

Hand in or read your answer aloud.